Samples and expectations

Observed sample values $x_1, \ldots, x_N$ are numbers, with $N \geq 1$. Expectations here require finite first absolute moments. Variance, covariance, and the associated transformation formulas require finite second moments. The sample mean is

$$\overline{x} = \frac{1}{N} \sum_{i=1}^{N} x_i.$$

A random variable's model mean is

$$\mu = \mathbb{E}[X].$$

For discrete values a_i with probabilities p_i,

$$\mathbb{E}[X] = \sum_{i=1}^{k} p_i a_i.$$

Continuous density f_X:

$$\mathbb{E}[X] = \int x f_X(x)\, dx.$$

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Samples and expectations (continued)

After observation, the sample mean is a numerical summary. Before observation, it is a random statistic. Expectation belongs to the probability model. Similar-looking formulas have different interpretations.

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Variance and the N−1 denominator

For fixed observed samples,

$$\overline{x} = \frac{1}{N}\sum_{i=1}^{N} x_i.$$

the usual unbiased sample variance is

$$s^2 = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \overline{x})^2.$$

The population variance is

$$\operatorname{Var}(X) = \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - \mu^2.$$

The $N-1$ denominator requires $N \geq 2$. For IID samples with finite variance, use $\sum_i (X_i - \overline{X})^2 = \sum_i (X_i - \mu)^2 - N(\overline{X} - \mu)^2$. Expectations give $N\sigma^2 - N(\sigma^2/N) = (N-1)\sigma^2$. Therefore

$$\mathbb{E}[s^2] = \sigma^2.$$

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Variance and the N−1 denominator (continued)

For the mean squared deviation of an entire finite population, division by N may instead be appropriate. Distinguish a descriptive statistic from an estimator.

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Means and variances under affine changes

For constants a, b,

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b,$$
$$\operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X).$$

A shift changes the mean but not spread. Negative scaling reverses direction while multiplying variance by the same positive factor a^2.

Independent X, Y:

$$\operatorname{Var}(X + Y) = \operatorname{Var} X + \operatorname{Var} Y.$$

Without independence, the general identity is

$$\operatorname{Var}(X + Y) = \operatorname{Var} X + \operatorname{Var} Y + 2 \operatorname{Cov}(X, Y).$$

Independence with finite second moments implies zero covariance. The converse generally fails.

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Densities, CDFs, and a uniform example

For an absolutely continuous random variable, the CDF has a derivative equal to its density almost everywhere. A continuous CDF alone does not guarantee a density. In the absolutely continuous case,

$$F_X(x) = \mathbb{P}(X \le x), \qquad f_X(x) = F_X'(x).$$

Interval probabilities satisfy

$$\mathbb{P}(a \le X \le b) = F_X(b) - F_X(a) = \int_a^b f_X(x)\,dx.$$

For $X \sim U(0, L)$, with $L > 0$, the density is $1/L$ on the interval. Its mean is

$$\mathbb{E}[X] = \frac{L}{2}.$$

Variance:

$$\operatorname{Var}(X) = \int_0^L \left(x - \frac{L}{2}\right)^2 \frac{1}{L}\,dx = \frac{L^2}{12}.$$

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Densities, CDFs, and a uniform example (continued)

For $L = 3$, the mean is 1.5 and variance is $3/4$. Translating the interval by a adds a to the mean and leaves variance unchanged.

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Gaussian standardization and the CLT

Standard normal:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

$X \sim N(\mu, \sigma^2)$:

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Standardization:

$$Z = \frac{X - \mu}{\sigma}$$

gives mean zero and variance one, requiring $\sigma > 0$.

For i.i.d. observations with $0 < \sigma^2 < \infty$, the central limit theorem states

$$\frac{\sqrt{N}(\overline{X} - \mu)}{\sigma} \xrightarrow{d} N(0, 1).$$

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Gaussian standardization and the CLT (continued)

This describes the limiting distribution of the standardized sample mean, not conversion of the raw data into Gaussian observations. Dependence or infinite-variance tails can violate these assumptions. The next section proves this IID finite-variance form using Lévy's continuity theorem.

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The CLT through independence and a second-order expansion

This proof uses Lévy's continuity theorem for the distributional convergence step; the general probability-theory proof of that theorem is outside this chapter. Standardize by setting $U_i = (X_i - \mu)/\sigma$. Then $\mathbb{E}[U_i] = 0$, $\operatorname{Var}(U_i) = 1$, and

$$Z_N = \frac{\sqrt{N}(\overline{X} - \mu)}{\sigma} = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} U_i.$$

Let $\varphi_N(t) = \mathbb{E}[e^{itZ_N}]$. Independence multiplies characteristic functions, and identical distributions make the factors equal.

$$\varphi_N(t) = \left[\varphi_U\left(\frac{t}{\sqrt{N}}\right)\right]^N.$$

A finite second moment gives a second-order expansion at zero.

$$\varphi_U(s) = 1 - \frac{1}{2}s^2 + o(s^2).$$

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The CLT through independence and a second-order expansion (continued)

No third moment is needed. Write $e^{iy} = 1 + iy - y^2/2 + R(y)$, with $R(y)/y^2 \to 0$ and a constant C such that $|R(y)| \le Cy^2$. Thus $R(sU)/s^2 \to 0$ almost surely and is dominated by integrable CU^2. Dominated convergence gives $\mathbb{E}[R(sU)] = o(s^2)$; linearity and $\mathbb{E}[U] = 0, \mathbb{E}[U^2] = 1$ give the expansion.

Consequently,

$$N \log \varphi_U(t/\sqrt{N}) = N \log \left(1 - \frac{t^2}{2N} + o(1/N)\right) \longrightarrow -\frac{t^2}{2}.$$

Hence $\varphi_N(t) \to e^{-t^2/2}$, the characteristic function of $N(0,1)$. Lévy's continuity theorem states that pointwise convergence to a characteristic function continuous at zero implies convergence in distribution. Thus Z_N converges to the standard normal. Uniqueness of characteristic functions alone would not establish convergence. The conclusion concerns standardized averages, not arbitrary large samples becoming normally distributed.

The CLT through independence and a second-order expansion (continued)

For fixed t and sufficiently large N, choose one logarithm branch near 1 and exponentiate the limit. Arbitrary principal complex logarithms do not satisfy an unrestricted product-addition rule. Finite-variance CLT gives no distribution-uniform finite-sample error guarantee.

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Standardized binomial distributions

Let B_i be independent fair 0/1 observations, with $S_N = \sum_i B_i$. Then

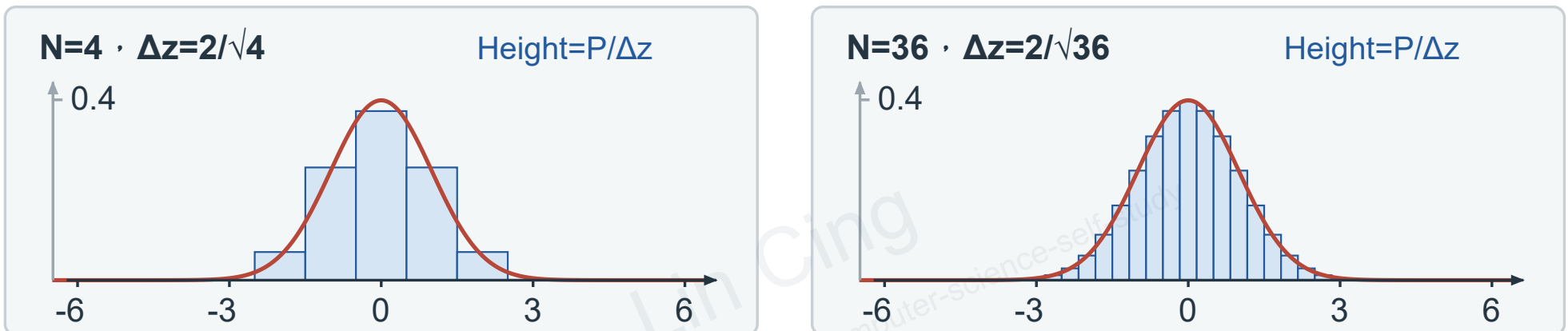
$$P(S_N = k) = \binom{N}{k} 2^{-N}, \quad \mathbb{E}[S_N] = N/2, \quad \mathrm{Var}(S_N) = N/4.$$

The standardized $Z_N = (S_N - N/2)/(\sqrt{N}/2)$ has spacing $\Delta z = 2/\sqrt{N}$. At $N = 4$, its values are $-2, -1, 0, 1, 2$, with probabilities $(1, 4, 6, 4, 1)/16$. Its mean is zero and variance one, but it remains discrete.

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Standardized binomial distributions (continued)

Standardized binomial: probability-area bars and N(0,1) density



Each bar has width Δz and height $P(S_N = k)/\Delta z$, so its area equals probability and total area is 1. The red curve is the standard normal density. Point probability itself cannot be compared with density height. These two finite examples illustrate the general theorem without proving it.

Error of a Monte Carlo average

For IID simulation outputs $Y_1, \dots, Y_N$ with mean μ and finite variance τ^2, the estimator $\widehat{\mu} = N^{-1} \sum_i Y_i$ satisfies

$$\mathbb{E}[\widehat{\mu}] = \mu, \qquad \mathrm{Var}(\widehat{\mu}) = \frac{\tau^2}{N}, \qquad P(|\widehat{\mu} - \mu| \geq \varepsilon) \leq \frac{\tau^2}{N\varepsilon^2}.$$

The last step is Chebyshev's inequality. Standard error scales as $N^{-1/2}$, a probabilistic sampling statement rather than a deterministic error bound on each run.

For $X \sim U(-1, 1)$, $Y = X^2$, integration gives $\mathbb{E}[Y] = 1/3$, $\mathbb{E}[Y^2] = 1/5$, hence $\tau^2 = 4/45$. At $N = 1000$, standard error is $\sqrt{4/45000} \approx 0.00943$, and $P(|\widehat{\mu} - 1/3| \geq 0.1) \leq 4/450 \approx 0.00889$. Correlated simulations require all cross-covariance terms; τ^2/N no longer applies unchanged.

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A joint probability table

For discrete variables, define joint mass

$$P_{ij} = \mathbb{P}(X = x_i, Y = y_j),$$
$$P_{ij} \geq 0, \qquad \sum_{i,j} P_{ij} = 1.$$

Marginals:

$$\mathbb{P}(X = x_i) = \sum_j P_{ij}, \qquad \mathbb{P}(Y = y_j) = \sum_i P_{ij}.$$

Independence requires every pair to satisfy

$$P_{ij} = \mathbb{P}(X = x_i)\mathbb{P}(Y = y_j).$$

Marginals alone generally do not determine the joint distribution. Means or one covariance value cannot establish independence either.

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A joint probability table (continued)

Joint probability table: marginals and covariance

X \ Y	Y=0	Y=1
X=0	0.2	0.1
X=1	0.3	0.4

PX=(0.3,0.7) **PY=(0.5,0.5)**

Total probability=0.2+0.1+0.3+0.4=1

E(X)=0.7 · E(Y)=0.5

E(XY)=0.4

Cov=0.4-0.7×0.5=0.05

VarX=0.21 · VarY=0.25

ρ=0.05/√0.0525=1/√21

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Conditional probability, Bayes, and zero correlation

If $P(X = x_i) = p_i > 0$, define

$$P(Y = y_j \mid X = x_i) = \frac{P_{ij}}{p_i}.$$

Using $P = \begin{bmatrix} 0.2 & 0.1 \\ 0.3 & 0.4 \end{bmatrix}$, with rows $X = 0, 1$ and columns $Y = 0, 1$,

$$P(Y = 1 \mid X = 0) = \frac{1}{3}, \quad P(Y = 1 \mid X = 1) = \frac{4}{7}.$$

total probability gives $P(Y = 1) = (1/3)(0.3) + (4/7)(0.7) = 0.5$. Bayes' formula reverses the conditioning.

$$P(X = 1 \mid Y = 1) = \frac{P(Y = 1 \mid X = 1)P(X = 1)}{P(Y = 1)} = \frac{(4/7)(0.7)}{0.5} = 0.8.$$

Zero covariance does not imply independence. Let X be uniform on $-1, 0, 1$ and $Y = X^2$. Then $\mathbb{E}[X] = \mathbb{E}[XY] = 0$, so $\mathrm{Cov}(X, Y) = 0$, but $P(Y = 0 \mid X = 0) = 1 \neq P(Y = 0) = 1/3$. Joint normality supplies the additional zero-covariance-implies-independence result.

Covariance and correlation

Correlation summarizes linear joint variation. Conditional probabilities from the joint distribution distinguish dependence from correlation.

$$\operatorname{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)].$$

For a discrete joint distribution,

$$\operatorname{Cov}(X, Y) = \sum_{i,j} P_{ij}(x_i - \mu_X)(y_j - \mu_Y).$$

Covariance depends on units; correlation removes scale.

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}.$$

With both variances positive,

$$-1 \leq \rho_{XY} \leq 1.$$

To prove the lower or upper bound, use

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Covariance and correlation (continued)

$$\operatorname{Var}\left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y}\right) = 2(1 - \rho_{XY}) \geq 0,$$

Replacing the minus by plus gives $1 + \rho_{XY} \geq 0$. Zero correlation generally does not imply independence.

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Covariance matrices and positive semidefiniteness

For a random column vector,

$$\mu = \mathbb{E}[X], \qquad V = \mathbb{E}[(X-\mu)(X-\mu)^T].$$

$V_{ij} = \sigma_{ij}$; the diagonal contains variances, and $V = V^T$.

For every coefficient vector c,

$$c^T V c = \operatorname{Var}(c^T X) = \mathbb{E}[(c^T(X-\mu))^2] \geq 0.$$

so V is positive semidefinite. A discrete joint distribution also gives the constructive factorization

$$V = \sum_{i,j} P_{ij} u_{ij} u_{ij}^T, \qquad u_{ij} = \begin{bmatrix} x_i \\ y_j \end{bmatrix} - \mu.$$

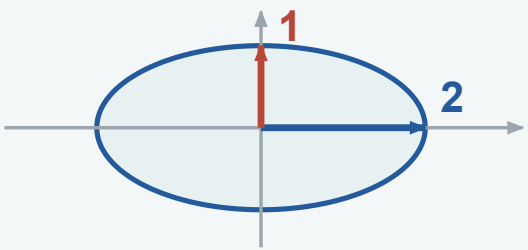
Positive definiteness means no nonzero combination $c^T X$ is almost surely constant. Deficient rank means an exact linear dependence among variable directions.

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Covariance matrices and positive semidefiniteness (continued)

Covariance geometry: V=diag(4,1), contour $x^TV^{-1}x=1$

$\lambda_1=4$, $\lambda_2=1$; semiaxes 2, 1

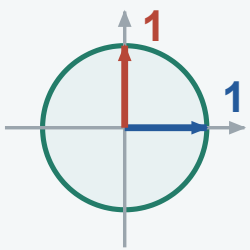


(2,0), (0,1): weighted squared length = 1

$J=e^TV^{-1}e$

Whitening: $z=(x_1/2,x_2)$

$J=\|z\|^2$



Equal scale: semiaxes (2,1) → radii (1,1)

The diagram uses a joint normal distribution with $V = \mathrm{diag}(4, 1)$, $\mu = 0$. Its equal-density contour $x^TV^{-1}x = 1$ is $x_1^2/4 + x_2^2 = 1$, with semiaxes 2 and 1. For general distributions, covariance alone need not determine elliptical density contours. Joint normality makes this quadratic form determine the density.

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Linear transformation of a random vector

Let

$$Y = AX + b.$$

Mean:

$$\mu_Y = A\mu_X + b.$$

Covariance:

$$V_Y = \mathbb{E}[(AX + b - \mu_Y)(\cdots)^T] = AV_XA^T.$$

The shift cancels under centering. The transformed covariance may be singular if the transformation has redundant output directions, such as repeated rows, or the original covariance already has null directions.

The variance of a^TX is a^TVa, and the covariance of a^TX, b^TX is a^TVb. Thus covariance defines a symmetric bilinear form on coefficient vectors.

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Sample covariance

Assume $n \geq 2$, and place samples in columns.

$$X_{\text{data}} \in \mathbb{R}^{M \times n}.$$

Row means:

$$\widehat{\mu}_j = \frac{1}{n} \sum_{k=1}^{n} (X_{\text{data}})_{jk}.$$

Center each row to obtain X_c. The usual sample covariance is

$$\widehat{V}_{n-1} = \frac{1}{n-1} X_c X_c^T.$$

Population/descriptive version:

$$\widehat{V}_n = \frac{1}{n} X_c X_c^T.$$

The two denominators give identical eigenvectors and eigenvalues differing by $n/(n-1)$. State the convention in a PCA report.

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Sample covariance (continued)

Unbiasedness requires $n \geq 2$, i.i.d. observations, and finite second moments. Centering gives $X_c \mathbf{1} = 0$, so rank is at most $\min(M, n-1)$. The nonzero covariance eigenvalues are scaled squared singular values, squaring the condition ratio on the nonzero spectrum. For deficient rank, the ordinary inverse condition number is already infinite. SVD avoids explicitly forming the Gram matrix.

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Whitening noise

For SPD error covariance V, V^{-1} weights directions inversely to their variances. Whitening gives

$$W = V^{-1/2}E.$$

Then

$$\mathrm{Cov}(W) = V^{-1/2}VV^{-1/2} = I.$$

The Mahalanobis length is

$$\|e\|_{V^{-1}}^2 = e^T V^{-1} e.$$

For diagonal V, divide each residual by its standard deviation. When covariance is estimated with more dimensions than samples, it can be singular; use shrinkage, dimension reduction, or a structural model.

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Multivariate Gaussian density

For $V \succ 0$, the density is

$$p(x) = \frac{1}{(2\pi)^{M/2}\det(V)^{1/2}} \exp\left[-\frac{1}{2}(x-\mu)^T V^{-1}(x-\mu)\right].$$

Its exponent has level sets

$$(x-\mu)^T V^{-1}(x-\mu) = r^2$$

which are ellipsoids for $r > 0$; for $r = 0$, the level set is the single point μ. Write the spectral decomposition

$$V = Q\Lambda Q^T, \qquad \Lambda = \operatorname{diag}(\lambda_i).$$

and set

$$Y = Q^T(X - \mu).$$

The transformed covariance is diagonal. For jointly Gaussian variables, the resulting uncorrelated components are independent. The Gaussian density factors in these coordinates, giving the statistical meaning of the spectral transformation.

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Gaussian normalization

Using

$$V = Q \Lambda Q^T$$

and orthogonality gives

$$(x-\mu)^T V^{-1} (x-\mu) = \sum_i \frac{y_i^2}{\lambda_i}, \qquad y = Q^T (x-\mu).$$

The orthogonal coordinate change has absolute determinant 1, so $dx = dy$. The positive integrand factors into one-dimensional Gaussian integrals.

$$\int_{\mathbb{R}^M} \exp\left(-\frac{1}{2} \sum_i \frac{y_i^2}{\lambda_i} \right) dy = (2\pi)^{M/2} \prod_i \lambda_i^{1/2}.$$

Also,

$$\prod_i \lambda_i^{1/2} = \det(V)^{1/2},$$

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Gaussian normalization (continued)

The one-dimensional constant follows by squaring the integral. If $I = \int_{\mathbb{R}} e^{-t^2/2} dt$, then polar coordinates give $I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2/2} r \, dr \, d\theta = 2\pi$, so $I = \sqrt{2\pi}$. Scaling each coordinate proves the stated normalization. A singular covariance instead gives a Gaussian supported on a lower-dimensional affine subspace, without an ordinary density relative to full-dimensional Lebesgue measure.

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Weighted least squares

For $A \in \mathbb{R}^{m \times n}$, $b, e \in \mathbb{R}^m$, $x \in \mathbb{R}^n$, use the model

$$b = Ax + e, \qquad \mathbb{E}[e] = 0, \qquad \mathrm{Cov}(e) = V \succ 0.$$

Ordinary least squares minimizes

$$\|b - Ax\|_2^2,$$

With fixed design and zero-mean errors, OLS can remain unbiased. Equal variances and uncorrelated errors, covariance $\sigma^2 I$, suffice for BLUE; independence is unnecessary. For general known SPD covariance, minimize the Mahalanobis residual.

$$J(x) = (b - Ax)^T V^{-1} (b - Ax).$$

The gradient is

$$\nabla J = -2A^T V^{-1}(b - Ax).$$

Setting it to zero gives the weighted normal equations.

$$A^T V^{-1} A x = A^T V^{-1} b.$$

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Weighted least squares (continued)

At full column rank,

$$\widehat{x} = (A^T V^{-1} A)^{-1} A^T V^{-1} b.$$

this stationary point is the unique global minimum. For nonzero h, expand

$$J(x+h) - J(x) = -2h^T A^T V^{-1}(b - Ax) + h^T A^T V^{-1} A h.$$

At $\widehat{x}$, the first term vanishes. Full column rank gives $Ah \neq 0$, and $V \succ 0$ gives $h^T A^T V^{-1} A h > 0$. Every nonzero displacement increases the objective.

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Three heteroscedastic measurements

Let $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$, $b = (2, 5, 4)^T$, $V = \operatorname{diag}(1, 4, 9)$. The whitened equations are

$$x_1 \approx 2, \quad \tfrac{1}{2}x_1 + \tfrac{1}{2}x_2 \approx \tfrac{5}{2}, \quad \tfrac{1}{3}x_2 \approx \tfrac{4}{3}.$$

Consequently,

$$A^T V^{-1} A = \begin{bmatrix} 5/4 & 1/4 \\ 1/4 & 13/36 \end{bmatrix}, \quad A^T V^{-1} b = \begin{bmatrix} 13/4 \\ 61/36 \end{bmatrix}.$$

Multiply the first normal equation by 4 to get $5x_1 + x_2 = 13$, and the second by 36 to get $9x_1 + 13x_2 = 61$. Substitute $x_2 = 13 - 5x_1$, obtaining $-56x_1 = -108$. Thus

$$\widehat{x} = (27/14, 47/14)^T, \qquad A\widehat{x} = (27/14, 37/7, 47/14)^T.$$

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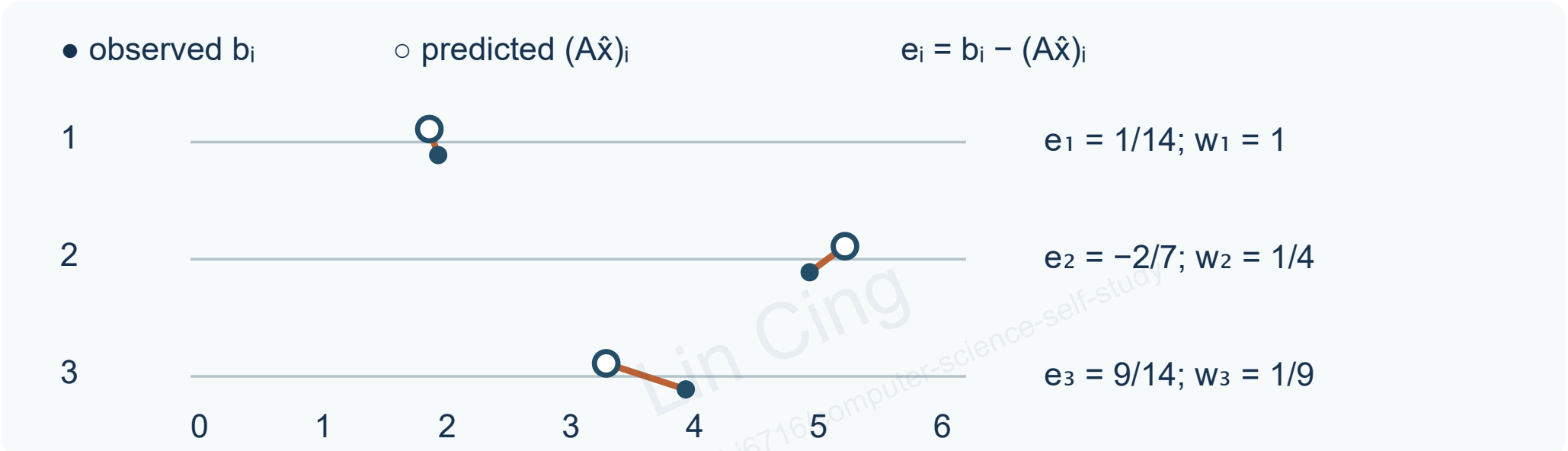
Three heteroscedastic measurements (continued)

Measurement	Residual $e_i = b_i - (A\widehat{x})_i$	Weighted square e_i^2/σ_i^2
1	$1/14$	$1/196$
2	$-2/7$	$4/196$
3	$9/14$	$9/196$

The sum is $J = 14/196 = 1/14$. Since $V^{-1}e = (1/14, -1/14, 1/14)^T$, $A^TV^{-1}e = 0$. The ordinary residual need not satisfy $A^Te = 0$; the whitened residual is orthogonal to the whitened design.

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Three measurements: predictions and weighted residuals



Filled circles are observations; open circles are predictions. Here $w_i = 1/\sigma_i^2$ and $e_i = b_i - (A\widehat{x})_i$. The weighted squared terms sum to $1/196 + 4/196 + 9/196 = 1/14$, and $A^T V^{-1} e = 0$.

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Ordinary and weighted least squares on the same data

Let $A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \\ 1 & 4 \end{bmatrix}$, $b = (4, 3, 4)^T$. Ordinary least squares gives $\widehat{x}_{OLS} = (-1/19, 49/38)^T$, with second residual $63/38$.

For fixed error covariance $V = \mathrm{diag}(1, 1/4, 1)$, use $W = V^{-1} = \mathrm{diag}(1, 4, 1)$. The weighted equations are

$$A^T W A = \begin{bmatrix} 6 & 1 \\ 1 & 21 \end{bmatrix}, \qquad A^T W b = (-4, 32)^T.$$

They give $\widehat{x}_W = (-116/125, 196/125)^T$.

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Comparing residuals: weights change the tradeoff

The weighted residual is $r_W = (84/25, 63/125, -168/125)^T$; its second component falls from $63/38$ to $63/125$. Verify column by column

$$A^T W r_W = 0, \qquad r_W^T W r_W = 1764/125.$$

The larger weight emphasizes the second measurement, while other residuals can increase. Weights require an error model or reliability justification; they do not improve every observation.

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Uncorrelated errors with unequal variances

For finite positive variances and mutually uncorrelated errors, with independence sufficient as well,

$$V = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_m^2),$$

minimize

$$J(x) = \sum_{i=1}^{m} \frac{(b_i - a_i^T x)^2}{\sigma_i^2}.$$

High-variance measurements receive less weight.

For repeated measurements of one scalar x,

$$A = (1, 1)^T, \qquad b = (b_1, b_2)^T.$$

the normal equation is

$$\left(\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}\right) x = \frac{b_1}{\sigma_1^2} + \frac{b_2}{\sigma_2^2}.$$

so

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Uncorrelated errors with unequal variances (continued)

$$\widehat{x} = \frac{b_1/\sigma_1^2 + b_2/\sigma_2^2}{1/\sigma_1^2 + 1/\sigma_2^2},$$

giving the precision-weighted average.

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Covariance of the estimator

The estimator is linear.

$$\widehat{x} = Gb, \qquad G = (A^T V^{-1} A)^{-1} A^T V^{-1}.$$

If b has covariance V,

$$\mathrm{Cov}(\widehat{x}) = GVG^T.$$

substituting G gives

$$\mathrm{Cov}(\widehat{x}) = (A^T V^{-1} A)^{-1}.$$

Diagonal entries are estimator variances; off-diagonal entries are covariances, not correlations until divided by the relevant standard deviations. The formula assumes fixed full-column-rank A and fixed known SPD V. Estimating V from the same data makes G random, so the plug-in inverse is not automatically the exact sampling covariance.

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Covariance of the estimator (continued)

For BLUE, $GA = I$ proves $\mathbb{E}[Gb] = x$. Any other linear estimator unbiased for every parameter is $Hb = (G + D)b$, with $DA = 0$. Since $GV = (A^T V^{-1} A)^{-1} A^T$, $GVD^T = 0$, hence $HVH^T - GVG^T = DVD^T \succeq 0$. Minimum covariance means PSD order, not entrywise order. Normality is unnecessary; exact confidence intervals and tests require additional distributional assumptions.

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Conditioning of weighted least squares

Avoid explicitly inverting V. For uncorrelated errors, whiten equation by equation.

$$\tilde{A} = V^{-1/2}A, \qquad \tilde{b} = V^{-1/2}b.$$

Then apply ordinary least squares to $\tilde{A}, \tilde{b}$.

Forming $\tilde{A}^T\tilde{A}$ squares the 2-norm condition number; inversion itself is not the source of the squaring. For general SPD $V = LL^T$, use triangular solves to form $\tilde{A} = L^{-1}A, \tilde{b} = L^{-1}b$, then QR or SVD. Large sparse problems may use matrix-free operations.

Shrinkage, regularization, or dimension reduction can handle uncertain or singular covariance estimates, but their changes to the statistical model must be stated.

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Recursive least squares with new observations

Let the old WLS estimate be $\widehat{x}_0$, with error covariance $W_0 \succ 0$. New data satisfy $b_1 = Hx + \eta$, with $\mathbb{E}[\eta] = 0$, $\mathrm{Cov}(\eta) = R \succ 0$, and new errors uncorrelated with the old data errors. Add the two weighted normal equations.

$$W_1^{-1} = W_0^{-1} + H^T R^{-1} H,$$
$$W_1^{-1}\widehat{x}_1 = W_0^{-1}\widehat{x}_0 + H^T R^{-1} b_1.$$

Subtract the first coefficient equation times $\widehat{x}_0$ from the second to obtain

$$\widehat{x}_1 = \widehat{x}_0 + K(b_1 - H\widehat{x}_0), \quad K = W_1 H^T R^{-1}.$$

The innovation $b_1 - H\widehat{x}_0$ is the difference between new observations and old predictions. To avoid repeated inversions, compute

$$S = R + HW_0H^T, \quad K = W_0H^T S^{-1}, \quad W_1 = W_0 - W_0H^T S^{-1} HW_0.$$

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Recursive least squares with new observations (continued)

To check the gain identity, $(W_0^{-1} + H^T R^{-1} H) W_0 H^T S^{-1} = H^T R^{-1} (R + H W_0 H^T) S^{-1} = H^T R^{-1}$. Multiplication by W_1 gives $K = W_1 H^T R^{-1}$. Multiplying the final covariance formula by W_1^{-1} verifies it similarly. The subtracted matrix is PSD, so $W_1 \preceq W_0$; under this model, new data do not increase linear-estimation uncertainty.

This is fixed-state recursive least squares and a simplified Kalman update. Dynamic filtering additionally needs state transitions, process errors, and a prediction step.

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Checking a recursive update numerically

Estimate a constant x. Old observation $b_0 = 10$ has variance 1, giving $\widehat{x}_0 = 10, W_0 = 1$. With $H = 1$, add two mutually uncorrelated new observations in sequence.

Step	New observation and variance	$S = R + W$	$K = W/S$	New estimate	New variance
1	$b_1 = 13, R_1 = 9$	10	$1/10$	$10 + (1/10)(3) = 103/10$	$9/10$
2	$b_2 = 9, R_2 = 1$	$19/10$	$9/19$	$103/10 + (9/19)(9 - 103/10) = 184/19$	$9/19$

The batch precision-weighted average is

$$\widehat{x} = \frac{10 + 13/9 + 9}{1 + 1/9 + 1} = \frac{184}{19}, \quad \operatorname{Var}(\widehat{x}) = \frac{1}{1 + 1/9 + 1} = \frac{9}{19}.$$

Both procedures agree exactly. A copy of an old observation has correlated error; treating it as independent counts the same information twice and understates uncertainty.

Robustness and model assumptions

WLS depends on its error model. Its assumptions include

- known or zero error means;
- known or credibly estimated covariance V;
- a correct linear conditional-mean model Ax;
- a joint covariance capturing dependence;
- full column rank for parameter identification.

Heavy tails without finite second moments invalidate covariance-based analysis. Outliers or omitted nonlinear terms can also make WLS unsuitable. Correlated errors are permitted when modeled correctly in V. An L_1 loss penalizes large residuals linearly and reduces sensitivity to vertical outliers, but does not automatically resist high-leverage design points. Linear least absolute deviations can be formulated as an LP.

Time-series errors often have nondiagonal covariance. Replacing it by $V = I$ changes both weights and uncertainty. Correlation is not a causal effect.

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Chapter review

- Distinguish sample summaries, random estimators, and distribution parameters; state assumptions for unbiasedness.
- Zero covariance does not generally imply independence; covariance matrices are positive semidefinite.
- Weighted least squares uses the error covariance; whitening gives an ordinary Euclidean projection.
- Recursive updates depend on assumptions about correlations between old and new errors.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 12.

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